



Improving Generalization of Genetic Programming for High-Dimensional Symbolic Regression with Shapley Value Based Feature Selection

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Received: 20 May 2024 / Revised: 2 October 2024 / Accepted: 11 November 2024 / Published online: 28 December 2024
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Abstract

Symbolic Regression (SR) on high-dimensional datasets often encounters significant challenges, resulting in models with poor generalization capabilities. While feature selection has the potential to enhance the generalization and learning performance in general, its application in Genetic Programming (GP) for high-dimensional SR remains a complex problem. Originating from game theory, the Shapley value is applied to additive feature attribution approaches where it distributes the difference between a model output and a baseline average across input variables. By providing an accurate assessment of each feature importance, the Shapley value offers a robust approach to select features. In this paper, we propose a novel feature selection method leveraging the Shapley value to identify and select important features in GP for high-dimensional SR. Through a series of experiments conducted on ten high-dimensional regression datasets, the results indicate that our algorithm surpasses standard GP and other GP-based feature selection methods in terms of learning and generalization performance on most datasets. Further analysis reveals that our algorithm generates more compact models, focusing on the inclusion of important features.

Keywords Feature selection · Generalization · Genetic programming · Symbolic regression

1 Introduction

In high-dimensional datasets, machine learning models face significant challenges, such as overfitting and high computational costs [1]. These challenges are particularly pronounced in symbolic regression (SR), which not only requires identifying optimal model coefficients but also more importantly discovering the model structure. In high-dimensional SR, traditional machine learning models often

struggle with overfitting, which hampers generalization ability. Although feature selection methods have shown promise in reducing dimensionality, their effectiveness is primarily demonstrated in classification tasks. Compared with classical regression, SR has a large and complex search space as the model structure can take various forms, which vastly increases the complexity of feature selection.

Genetic Programming (GP), an evolutionary computation method, is well-suited for SR as it generates optimal solutions without requiring prior knowledge of the solution's form or size [2]. However, while GP inherently offers feature selection capabilities through its (tree-based) model construction, it often struggles with robustness, especially when dealing with high-dimensional datasets. This is because higher dimensions increase the risk of overfitting, making it more difficult for GP to construct effective and generalizable models.

Feature selection offers a potential solution by reducing model complexity through the selection of a subset of important features [3]. This not only speeds up the learning process but also improves the performance of models on unseen data. Despite its importance, there has been limited research on

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explicitly integrating feature selection with GP for SR. One example is GP with permutation importance for high-dimensional SR [4]. However, the permutation importance method has limitations in identifying correlated features because it assesses the influence of each feature independently on the model's effectiveness. In the situation where there are both anomalous data instances and correlated features, permutation importance cannot accurately represent the true importance of a feature [5].

To address the limitations of permutation importance, this work utilizes the Shapley value, which originates from cooperative game theory and fairly and uniquely distributes outcomes based on individual contributions. In feature selection, the Shapley value assesses each feature's impact on model performance by considering the marginal contribution of each feature in combination with others. Building on this, we propose a new Shapley value-based feature selection method for high-dimensional SR in GP. Specifically, it uses a two stage framework where the first stage selects important features using the Shapley value and the second stage uses these selected features for GPSR. The primary contributions of this paper are as follows:

1. This paper proposes a new GP-based feature selection with Shapley value to effectively identify important features, helping to mitigate overfitting in high-dimensional SR.
2. This paper employs a threshold-based approach to select top $\log_2(D)$ features, resulting in more compact models. This method balances model complexity with training and test performances by retaining only the most important features.
3. Extensive experiments are conducted to demonstrate the effectiveness of the proposed method in high-dimensional SR. The results show that the method not only enhances the learning process by selecting important features but also significantly improves generalization performance on unseen data.

The following sections outline the structure of this paper. In Sect. 2, we provide background information, including a concise overview of GPSR and a review of existing contributions related to feature selection and generalization in GP. Section 3 introduces our novel method introducing Shapley value for feature selection in GPSR. Section 4 details the experimental design, encompassing ten high-dimensional regression datasets used to test among the proposed method, standard GP and other GP-based feature selection methods. This section

also covers the creation of training and test sets, along with parameter settings. Section 5 presents and analyzes the performance of proposed method and other GP-based methods on both training and test sets on the ten datasets. Section 6 delves into a further analysis, including the outcomes of the feature selection process, the details of the evolved models, the average size and number of features incorporated within these models, as well as the computational cost. Finally, in Sect. 7, we draw conclusions from this work and outline important directions for future research.

2 Background

2.1 Genetic Programming and Symbolic Regression

Genetic Programming (GP) is an evolutionary computation technique designed to create computer programs to solve problems at hand [5]. As shown in Fig. 1, GP begins with generating an initial population of random programs. The population is then evolved through iterative cycles of evaluation, parent selection, and offspring generation by crossover and mutation operators. This evolutionary cycle continues until a predefined termination criterion is met. The output is the program with the best fitness, representing the optimal solution to the problem.

Symbolic Regression (SR) is a form of regression analysis with function identification and coefficient fitting [4]. In contrast to classical regression, SR has the unique ability to produce a mathematical model without pre-existing knowledge on the model structure and dimensions [6]. The SR problem can be formalized as follows: Given a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where $\mathbf{x}_i \in \mathbb{R}^d$ represents the input feature vector for the i^{th} sample, and $y_i \in \mathbb{R}$ represents the corresponding output, the goal of SR is to discover a mathematical expression $f(\mathbf{x})$ that approximates the underlying relationship between the inputs and the output: $y_i = f(\mathbf{x}_i) + \epsilon_i$ where ϵ_i represents the noise or error term, typically assumed to follow some distribution. The problem can be formulated as a learning problem: $\min_{f \in \mathcal{F}} \mathcal{L}(f(\mathbf{x}), y)$ where $\mathcal{F}$ is the space of all possible mathematical expressions, $\mathcal{L}(f(\mathbf{x}), y)$ is a loss function that measures the difference between the predicted values $f(\mathbf{x})$ and the true target values, such as Mean Squared Error (MSE).

This flexibility renders GP highly appropriate for SR, as it can adaptively find optimal solutions without predefined model structures or data distributions. Numerous researchers have proposed various GP methods to improve the performance and efficacy of SR [7, 8]. However, on high-dimensional datasets,

Fig. 1 The evolutionary process of the GP system



the vast number of features exponentially increases the complexity of finding accurate symbolic models. GP may encounter difficulty in distinguishing important features from irrelevant or redundant ones, which leads to overfitting and weak generalization. Improving the generalization ability of GP for Symbolic Regression (GPSR) remains an open challenge.

2.2 Genetic Programming for Feature Selection

Feature selection is able to improve the learning ability by reducing irrelevant and redundant features. Feature selection methods can be classified into three categories, i.e., filter methods, wrapper methods and embedded methods [9]. Filter methods, often serving as a preprocessing technique, operate independently of any learning algorithm, selecting feature subsets according to various standards, e.g., correlation coefficient and mutual information [10]. Wrapper methods select feature subsets based on the performance of a learning algorithm with the candidate feature subsets [11] and embedded methods combine the feature selection with the learning process [12, 13].

GP, as an embedded method, inherently has the ability to navigate the feature space and identify important features, making it a favorable choice for feature selection [4, 14]. However, this inherent capability is less effective when dealing with high-dimensional datasets [5]. Consequently, various existing studies have investigated methods aimed at enhancing the effectiveness of GP in selecting features. For instance, Sandin et al. [15] proposed a GP based method specifically designed for unbalanced, high-dimensional datasets. The method explores diverse combinations of feature selection metrics, including information gain, χ^2 , and odds ratio, aiming to identify an unbiased estimator that accurately assesses the discriminative capacity of features. Chen et al. [14] proposed GP with Feature Selection (GPWFS) for high-dimensional symbolic regression, aiming to enhance generalization capability. GPWFS divides the evolutionary process into two stages: A feature selection stage where unique features from good individuals are collected, forming a subset of the original features, and a standard GP evolutionary process with population reinitialization using features from the selected subset. Based on GPWFS, Chen et al. [4] later presented a permutation importance-based feature selection method for high-dimensional GPSR. This method obtains feature importance by shuffling values in the learned model, contributing to improved generalization of GPSR. Dick [16] proposed a GP-based method using a sensitivity-like analysis method to identify important features. This method observes changes in error rates to calculate feature importance. Moreover, feature importance can be incorporated back into GP, influencing feature selection during the tree construction process.

2.3 Generalization in GP for Symbolic Regression

Generalization is vital for a learned model to make accurate predictions on unseen data. It serves as a fundamental performance metric for supervised learning algorithms, validating the effectiveness of the learning process. Overfitting is the opposite of generalization, characterized by a low training error but a high test error. Models with strong generalization capabilities perform well not only on training data, but more importantly, on unseen data. However, learning models from a high-dimensional feature space poses a risk of overfitting, resulting in poor generalization [4].

Recently, there has been a heightened focus on improving generalization in GPSR. Chen et al. [17] proposed a multi-objective approach to enhance generalization in GP. They consider training error and Rademacher complexity as the two objectives, with Rademacher complexity computed based on the maximum correlation between the model and Rademacher variables on training instances. Zhang et al. [7] proposed a modular and hierarchical multi-tree GP framework designed to construct high-level features that enhance the generalization of GP models. The approach incorporates a macro parent-repair strategy to eliminate redundant and irrelevant features, along with a macro crossover operator that helps preserve interactive features between them. The empirical results demonstrate that the modular representation not only improves generalization but also reduces the model size. Munoz et al. [8] explored transfer learning in GP to enhance its generalization capabilities. Their research particularly focuses on the effectiveness of transferred solutions when the source and target tasks belong to different domains and lack a common feature space.

In summary, the existing literature primarily addresses the enhancement of regression performance and generalization within GP frameworks, but these improvements applied in high-dimensional datasets are limited.

3 Proposed Feature Selection Method

3.1 Overall Structure

This paper proposes a novel feature selection method termed GP with Shapley value (GPSHAP) which is classified as the embedded algorithm. This approach uses the capability of GP to explore the search space for automatic feature selection [14]. The underlying assumption is that individuals possessing good fitness likely contain important features. However, not all features within these individuals are important and some may be redundant. Consequently, these features constitute a candidate subset of potentially important features. By assessing the importance of these candidates, we can identify those that are important.

Figure 2 illustrates the data flow diagram of the proposed GP system, including our new feature selection component GPSHAP and the standard GPSR framework. The training process of the system involves two phases. In the initial stage, GPSHAP is deployed to select a subset of crucial features from the training data. Subsequently, standard GPSR is applied to the training data, utilizing only these selected features. The details of GPSHAP will be presented in Sects. 3.3.

3.2 Fitness Function

In this paper, the normalized root mean square error (NRMSE) is utilized as the fitness function to evaluate individual performance in both feature selection and regression. The NRMSE is defined by the following equation:

$$NRMSE = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (f(X_i) - Y_i)^2}}{Y_{max} - Y_{min}} \quad (1)$$

where N represents the number of instances, $f(X_i)$ denotes the outputs generated by the model for the i^{th} instance and Y_i corresponds to the target output for the i^{th} instance. $Y_{max} - Y_{min}$ signifies the range of the target output on the training dataset.

3.3 Feature Selection based on Shapley Value

As highlighted earlier, not all features presented in good individuals are important. Therefore, a vital aspect of the feature selection method is the concept of feature importance which is a quantitative measure of the significance of the features. Feature importance measures how closely linked a feature is to the target feature or how much it helps reduce the difference between the predictions and the target values [4].

The Shapley value, originating from cooperative game theory, represents a concept of fairly distributing the total game gain among game players. As an additive feature attribution algorithm, it adheres to four key mathematical axioms, i.e., efficiency, symmetry, dummy, and additivity [5, 18]. Efficiency ensures that the total contributions match the

difference between the model's prediction and the average prediction. Symmetry assigns equal Shapley value to equally contributing features. Dummy gives zero value to any feature that does not enhance model performance. Additivity allows the Shapley value of a feature which can be aggregated across models. These axioms enable the Shapley value to impartially attribute contributions to each feature. Features that consistently improve model performance across various feature subsets will accrue higher Shapley values. Conversely, features that seldom enhance performance or only do in limited feature subsets will accumulate lower Shapley values. Therefore, more important features possess higher Shapley values, indicating a greater contribution to the target values.

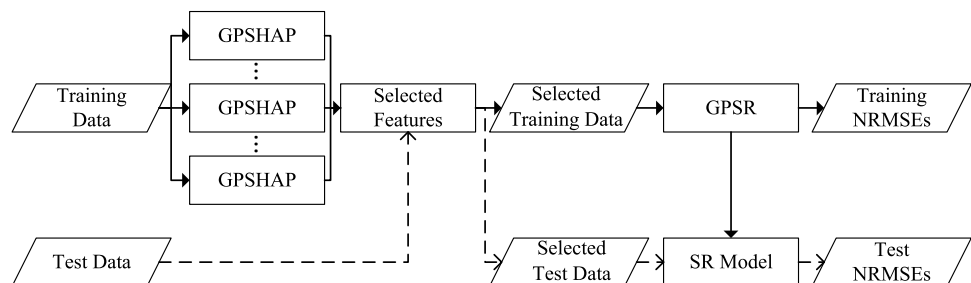
To employ Shapley value for quantifying feature importance in GPSR, we consider the prediction outcome of each individual as the gain from a cooperative game, where different features within the individual are viewed as players in the game. Let $F = \{v_1, \dots, v_k\}$ represent the k unique features in the individual. S denotes a subset of F , i.e., $S \subseteq F$. Additionally, $f_{S \cup \{i\}}$ refers to a GP model trained with the i^{th} feature, whereas f_S denotes a model trained without the i^{th} feature. Therefore, the difference in the predictions can be expressed as $f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)$, where x_S represents the value of the features in subset S . The Shapley value, capturing the contribution of each feature towards the prediction outcome, takes into account all possible combinations of features and is formally defined in Eq. (2).

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)] \quad (2)$$

where $|F|$ denotes the number of features and $|S|$ represents the number of features in the subset S . $\frac{|S|!(|F| - |S| - 1)!}{|F|!}$ calculates the weight of the contribution of feature i when it is added to subset S .

Let illustrate the calculation of the Shapley value with a simplified example involving a model with only two features, i.e., $F = \{1, 2\}$ and compute the Shapely value of Feature 1, i.e., ϕ_1 . The feature subsets that do not include Feature 1 are $\{\}$ and $\{2\}$. When Feature 1 is added to $\{\}$, the weight is equal to $\frac{0!(2-0-1)!}{2!} = \frac{1}{2}$ and the marginal contribution is $f_{\{1\}}(x_{\{1\}}) - f_{\{\}}(x_{\{\}})$. When

Fig. 2 The data flowchart of the proposed GP system



Feature 1 is added to $\{2\}$, the weight is $\frac{1!(2-1-1)!}{2!} = \frac{1}{2}$ and the marginal contribution is $f_{\{1,2\}}(x_{\{1,2\}}) - f_{\{2\}}(x_{\{2\}})$. So $\phi_1 = \frac{1}{2}[f_{\{1\}}(x_{\{1\}}) - f_{\{\}}(x_{\{\}})] + \frac{1}{2}[f_{\{1,2\}}(x_{\{1,2\}}) - f_{\{2\}}(x_{\{2\}})]$.

However, calculating the Shapley value for high-dimensional data is time-consuming due to the need to enumerate all possible combinations of features. To tackle this challenge and improve computational efficiency, this study adopts a sampling-based approximation method for ϕ_i , as introduced in [19]. This approximation approach computes ϕ_i by calculating the average marginal contribution of a feature over multiple sampling iterations, detailed in Eq. (3). While Eq. (2) considers all possible combinations of features, Eq. (3) considers only a subset of these combinations by random sampling. This approach allows for a more computationally efficient estimation of feature contributions, especially on high-dimensional datasets where evaluating all combinations becomes impractical.

$$\phi_i = \frac{1}{T} \sum_{t=1}^T (f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)) \quad (3)$$

where T is the total number of sampling steps.

Figure 3 is the flowchart of GPSHAP. The whole process is described as follows.

1. Conduct n parallel GP runs, each producing a best-of-run individual I_{b_i} . The requirement $n \geq 30$ aims to minimize the impact of random seed bias in GP runs on the provided training data.
2. Collect terminal nodes from each I_{b_i} to form the unique feature subset UF_i .
3. Compute Shapley values for each feature ($Score_i(I_{b_i})$) in UF_i using Eq. (3).

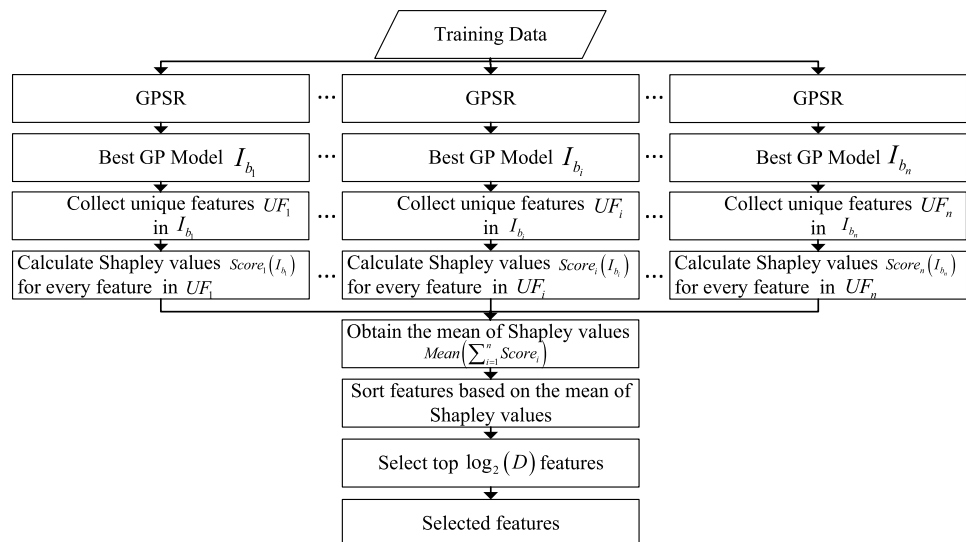
4. Aggregate the Shapley values from the n parallel runs by calculating the mean denoted as $Mean(\sum_{i=1}^n Score_i(I_{b_i}))$. This mean value represents as the feature importance on the training data. If a feature is absent in any best-of-run individuals, its Shapley value is set to 0, indicating a negligible contribution to the overall prediction outcome.
5. Rank features based on the mean of Shapley values.
6. Building upon the established previous work on determining an appropriate threshold [20], we select the top $\log_2(D)$ features as the final set of features where D represents the total number of available features in the dataset.

Our new GPSR method utilizes a subset of features selected by GPSHAP, offering a selection metric that is problem-independent. This characteristic is particularly well-suited for regression problems lacking domain knowledge as it does not rely on specific problem domains.

4 Experiment Design

We conducted a series of experiments to show the capability of GPSHAP in selecting features and its potential on enhancing the generalization performance of GPSR. GPSHAP has been compared against GPSR coupled with different feature selection algorithms aiming to evaluate the advantages and unique capabilities of our proposed method.

Fig. 3 The flowchart of GPSHAP



4.1 Benchmark Methods

This study conducts a comparison of GPSHAP against three benchmark approaches that incorporate feature selection into GPSR. Brief descriptions of the three benchmarks are provided below.

1. GPRF represents a GPSR method that employs Random Forests (RFs) for feature selection. To account for the impact of random seeds on the permutation feature within RF, 30 runs are performed with distinct seed values. This approach helps mitigate any potential bias arising from the stochastic nature of RFs, thus ensuring a more reliable and accurate assessment of feature importance.
2. GPWFS represents a feature selection method within GPSR that simultaneously considers the feature frequency and regression performance. A brief overview of GPWFS is provided in Sect. 2.2. For an in-depth understanding, interested readers are encouraged to consult reference [14].
3. GPPI refers to a GPSR method that utilizes permutation importance for feature selection. A brief overview of GPPI can be found in Sect. 2.2. For comprehensive insights, interested readers are encouraged to consult the reference [4].

We performed an ablation study to investigate the effects of the Shapley value and different thresholds for selecting important features. In GPSHAP_50%, the Shapley value is used to calculate feature importance, and the top 50% of features are selected based on this importance. Similarly, GPSHAP_70% selects the top 70% of features. The comparison between GP and GPSHAP evaluates the impact of incorporating Shapley values, while the comparisons between GPSHAP and the threshold-based variants (GPSHAP_50% and GPSHAP_70%) assess the effectiveness of different feature selection thresholds.

Beyond the aforementioned feature selection techniques, standard GP, which employs the complete feature set and include built-in feature selection, is employed as a baseline for comparison. The primary focus of this study lies in assessing the performance of GP when combined with different feature selection methods. All GP methods have been developed by the *deap* package [21]. For feature selection, RF is executed the “RandomForestRegressor” function within the *scikit-learn* package [22]. We use the “SHAP.SamplingExplainer” function from the SHAP package to compute Shapley values [19].

Table 1 Parameter settings

Parameter	Values
Population size	512
Generation for feature selection	50
Generation for symbolic regression	50
Crossover rate	0.9
Mutation rate	0.1
Elitism	5
Maximum tree depth	10
Initialization	Ramped-half&Half
Minimum initialization depth	2
Maximum initialization depth	6
Function set	$+$, $-$, $*$, $Inv(\frac{1}{x})$, $\sqrt{x}$
Terminal set	Features (Selected features) Random constant $\in [-1.0, 1.0]$
Fitness function	NRMSE

4.2 Parameters

For an accurate comparison, the parameter settings in GPSHAP, standard GP, GPPI, GPRF and GPWFS are kept consistent, as summarized in Table 1. For each iteration of RF-based feature selection, a 100-tree forest is constructed, adhering to the default configuration in the *scikit-learn* package. The feature selection criterion in RF is the same as that of GPPI, emphasizing features with positive importance values. In GPWFS, the good individuals refer to the top 10% individuals, resulting in a selection of 51 individuals. For all SHAP-based GP methods, the number of feature subsets is set to $2 \times |UF| + 2048$, following the default setting of the “SHAP.SamplingExplainer” function [19] where $|UF|$ is the number of unique features in the best-of-run GP individual.

4.3 Datasets

The two synthetic datasets presented in Table 2 include F_1 and F_2 . F_1 is derived from Newton’s law of gravitation, where g represents the gravitational constant assigned a value of $6.67408E - 11$. F_2 is adopted from reference [23]. Furthermore, to evaluate the feature selection capabilities of the methods under consideration, noisy features have been introduced into each dataset enabling a comprehensive assessment of the methods’ ability to identify and handle irrelevant or misleading input variables.

Eight real-world datasets are used in this work. Concerning the two datasets sourced from the Penn Machine Learning Benchmarks (PMLB) [24], they are identified as 4544_GeographicalOriginOfMusic (4544_GOM) and 505_tecator, from which variables contributing to imbalance have been omitted. These datasets are specifically curated for

Table 2 Two synthetic functions

Functions	Training samples	Test sample	Noise
$F_1 = -g \frac{X_1 X_2}{X_3^2}$	70 points $X_1, X_2 = \text{rnd}[0, 1]$ $X_3 = \text{rnd}[1, 2]$	30 points $X_1, X_2 = \text{rnd}[0, 1]$ $X_3 = \text{rnd}[1, 2]$	50 input variables $= \text{rnd}[0, 1]$
$F_2 = \frac{30X_1 X_3}{(X_1-10)X_2^2}$	1000 points $X_1, X_3 = \text{rnd}(-1, 1)$ $X_2 = \text{rnd}(1, 2)$	10000 points $X_1, X_3 = \text{rnd}(-1, 1)$ $X_2 = \text{rnd}(1, 2)$	50 input variables $= \text{rnd}[0, 1]$

Table 3 Benchmark problems

Name	#Features	#Total instances	#Training instances	#Test instances
F_1	53	100	70	30
F_2	53	11000	1000	10000
CCN	122	1994	1395	599
CCUN	124	1994	1395	599
4544_GOM	117	1059	767	328
505_tecator	124	240	168	72
QT-1	1024	427	299	128
QT-2	1024	395	277	118
QT-3	1024	692	484	208
QT-4	1024	389	272	117

regression tasks. Secondly, from the UCI Machine Learning Repository, two datasets have been selected: The Communities and Crime Unnormalized dataset (CCUN) [25] and the Communities and Crime Normalized dataset (CCN) [26]. Instances with missing values have been removed, resulting in a reduced number of instances compared to the original datasets. Lastly, four datasets from OpenML [27] are employed, designated as QSAR-TID-10878 (QT-1),

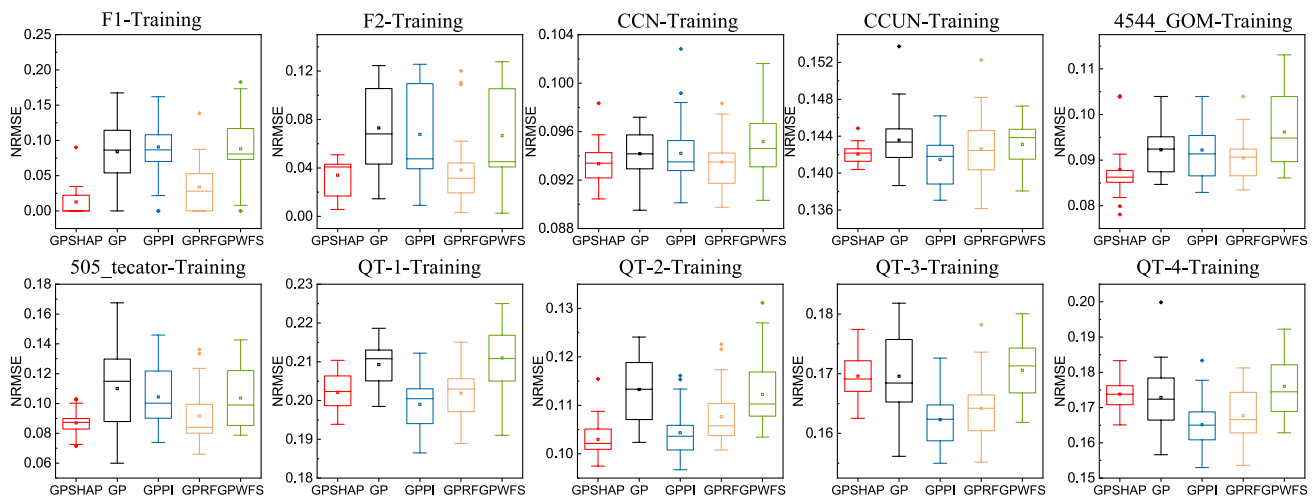
QSAR-TID-12415 (QT-2), QSAR-TID-12514 (QT-3), and QSAR-TID-12949 (QT-4). These datasets comprise QSAR data, indicating the activity levels of various compounds on drug targets.

Table 3 provides a comprehensive overview of the ten datasets utilized in this study. This research employs a widely-recognized partitioning strategy in machine learning [28], which involves randomly split the data, using 70% for training and 30% for testing. Throughout the feature selection process, all methods exclusively utilize the training sets. To ensure the robustness and reliability of the results, we run each method 30 times on each dataset independently.

5 Experiment Results

This section will show a comparative analysis among GPSHAP, GPPI, GPRF and GPWFS, with Standard GP serving as the baseline. The primary goal of this comparison is to evaluate the impact of the four feature selection methods on the learning efficiency and generalization capability of GPSR.

Figures 4 and 7 illustrate the distribution of Normalized Root Mean Square Errors (NRMSEs) for the best-of-run

**Fig. 4** The distributions of **training** NRMSEs of the 30 best-of-run individuals

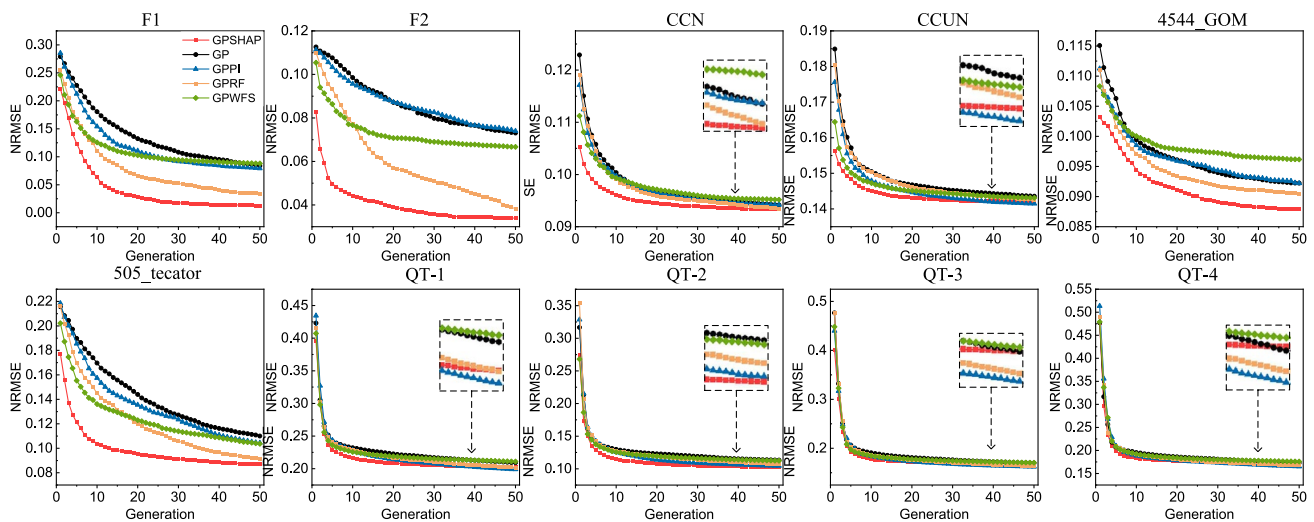


Fig. 5 The evolutionary curves on the mean NRMSE of 30 best-of-generation individuals on the **Training Sets**

individuals from 30 GP runs on the training and test sets, respectively. Furthermore, Figs. 5 and 8 display the evolutionary curves of training errors and the corresponding test errors from these 30 GP runs. At every generation, the minimal NRMSEs on the training set are recorded and we also report the test NRMSEs of the recorded best-of-generation GPSR models. It is important to note that test NRMSEs are tracked to assess generalization capabilities of best-of-generation GPSR models during the evolutionary process, but they are not utilized during this process.

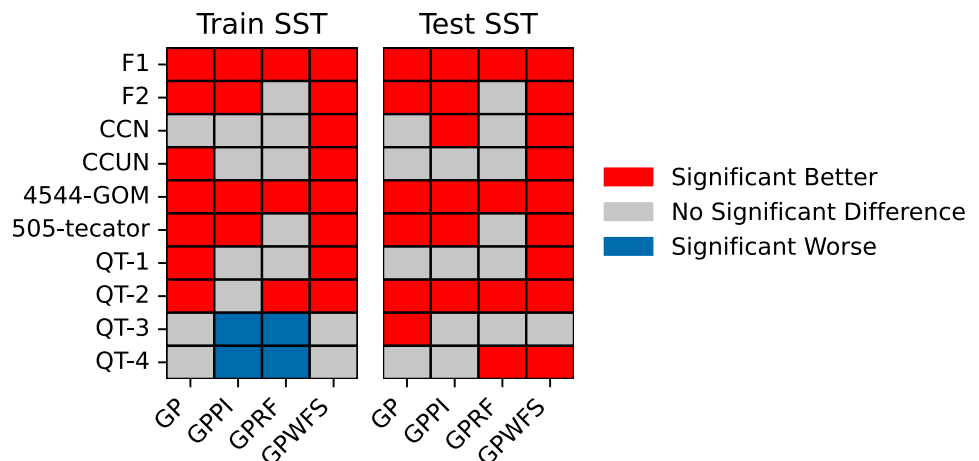
Figure 6 employs the Mann–Whitney U Test to analyze the training and test NRMSEs of the 30 best-of-run models for GPSHAP, standard GP, GPPI, GPRF and GPWFS. The threshold for statistical significance is set at 0.05. The red squares indicate that GPSHAP significantly outperforms the comparison methods, the grey squares signify no significant difference, and the blue squares indicate that GPSHAP is significantly inferior to the comparison methods.

5.1 Results on the Training Sets — Learning Ability

This paper investigates the influence of feature selection methods on the learning performance of GP when applied to high-dimensional regression datasets. To assess regression performance, we employ the training NRMSEs as a quantitative measure. This analysis enables us to evaluate the effectiveness of various feature selection methods in optimizing GP’s learning capabilities and achieving accurate predictions for high-dimensional SR.

Overall, Fig. 6 shows that GPSHAP is statistically superior or has similar training performance to the other methods on eight of the ten datasets. Specifically, as depicted in Fig. 4, GPSHAP exhibits superior training performance compared to other four GPSR methods on four of the ten training sets, i.e., F_1 , CCN, 4544_GOM, QT-2. Although the boxplot ranges of GPSHAP and the other four methods show some overlap for these datasets, their median values

Fig. 6 Statistical significant results on **training** and **test** sets



are distinct, indicating that GPSHAP substantially outperforms the others in terms of training performance on these datasets. These discrepancies can be attributed to the capability of Shapley value to accurately identify feature importance. On F_2 and 505_tecator, while GPSHAP significantly outperforms three of the four comparison algorithms, its performance does not show a statistically significant difference when compared to GPRF.

Figure 5 illustrates the evolutionary curves of GPSHAP, GP, GPPI, GPRF and GPWFS on training datasets, revealing that GPSHAP converge more rapidly than the other methods on all datasets. The superior initial performance of GPSHAP can be attributed to its feature selection strategy, which prioritizes the top $\log_2(D)$ important features. On F_1 , F_2 , 4544_GOM, and 505_tecator, GPSHAP consistently outperforms the other methods during the entire evolutionary process, displaying notably superior performance. While on QT-1, QT-3 and QT-4, the final outcomes of GPSHAP do not surpass those of GPPI and GPRF, it exhibits more favorable performance in the initial stages of evolution and achieves quicker convergence compared to them. This indicates that, GPSHAP has the ability to rapidly search effective solutions during the evolutionary process, despite not achieving the best final performance on these datasets. This improvement likely stems from the reduction in feature space afforded by feature selection, which consequently narrows the search space for GP, facilitating the discovery of superior models.

5.2 Results on the Test Sets — Generalization Ability

Overall, GPSHAP consistently performs better or similar than other GP methods on the test sets, as shown in Fig. 6.

The performance trends observed on the test sets closely mirror those seen on the training sets. Specifically, as illustrated in Fig. 7, GPSHAP exhibits lower test errors compared to GP and GPWFS on all ten datasets, underscoring its superior generalization capabilities on these datasets. Despite GPSHAP inferior training performance relative to GP and GPWFS on QT-3 and QT-4, it demonstrates better generalization on these two datasets. Furthermore, GPSHAP significantly outperforms GPPI and GPRF on six out of the ten test sets.

On F_1 , GPSHAP demonstrates markedly superior generalization performance compared to the other four methods, highlighting the efficacy of the Shapley value in pinpointing informative features and retaining pertinent ones during the evolutionary process. On F_2 , GPSHAP significantly outperforms the other three algorithms, except GPRF. The results indicate that GPSHAP excels in uncovering the relationships between input variables and target variables on synthetic datasets.

On the eight real-world datasets, the generalization capabilities of GPSHAP surpass those of the other four methods on four of the eight datasets, i.e., CCN, 4544_GOM, 505_tecator and QT-2. As evidenced by the evolutionary curves in Fig. 8, GPSHAP, GPWFS and GP exhibit tendencies towards overfitting on CCN, as indicated by an increase in test error towards the end of the evolutionary process. However, the degree of overfitting of GPSHAP is considerably less than that observed with GP and GPWFS. Despite GPSHAP slightly inferior performance compared to GPPI on CCUN, QT-3 and QT-4, the difference is not statistically significant, as corroborated by the significance tests presented in Fig. 6. On 4544_GOM, GPSHAP demonstrates more pronounced fluctuations than other four methods, yet it exhibits superior learning efficiency. In the later stages of

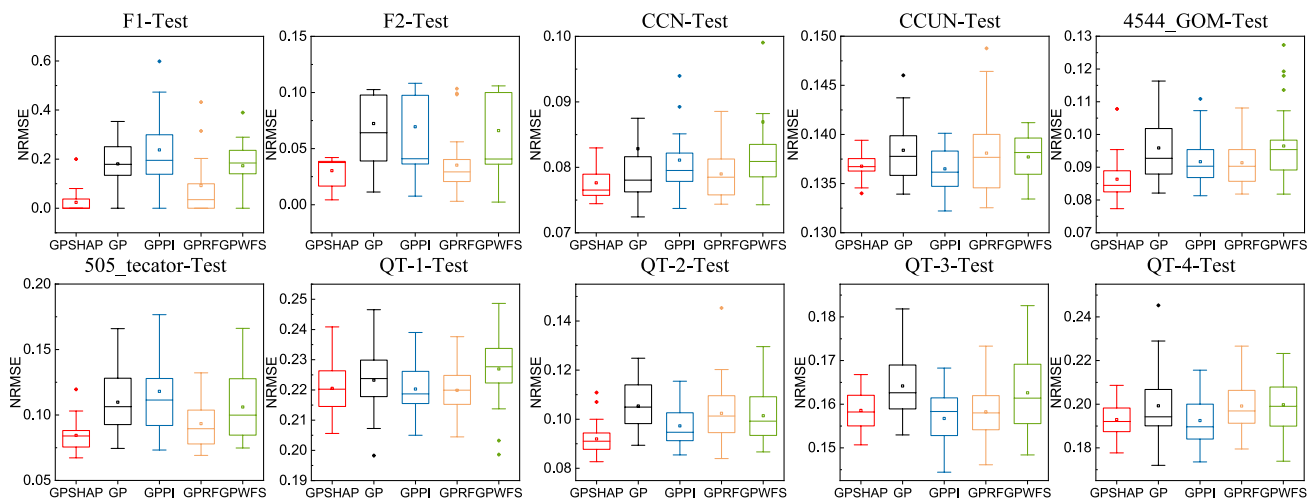


Fig. 7 The distributions of test NRMSEs of the 30 best-of-run individuals

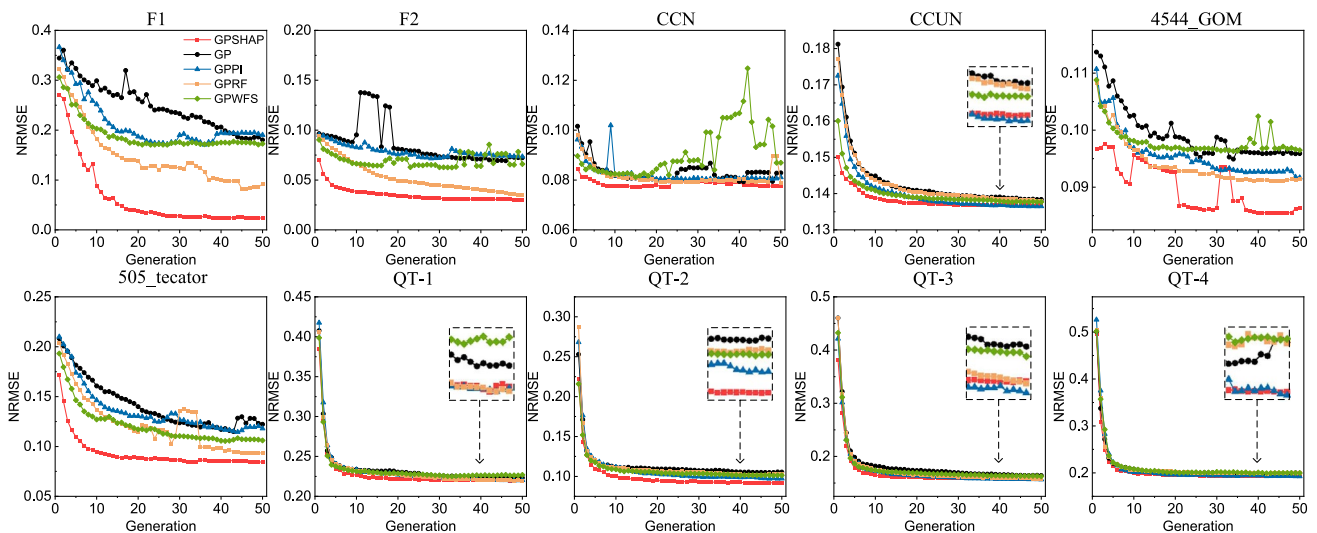


Fig. 8 The evolutionary curves of the mean NRMSE of 30 best-of-generation individuals during every generation on the **Test Sets**

Table 4 Statistical comparisons on the **training** NRMSE score for GP methods on all datasets. (“+”, “=” and “−” indicate that method in the row is better than, similar to or worse than the method in the column)

	GPSHAP_50%	GPSHAP_70%	GP
GPSHAP	1(+)/4(=)/5(−)	3(+)/4(=)/3(−)	7(+)/3(=)/0(−)
GPSHAP_50%	—	2(+)/9(=)/0(−)	10(+)/0(=)/0(−)
GPSHAP_70%	—	—	9(+)/1(=)/0(−)

Table 5 Statistical comparisons on the **test** NRMSE score for GP methods on all datasets. (“+”, “=” and “−” indicate that method in the row is better than, similar to or worse than the method in the column)

	GPSHAP_50%	GPSHAP_70%	GP
GPSHAP	3(+)/4(=)/3(−)	3(+)/6(=)/1(−)	6(+)/4(=)/0(−)
GPSHAP_50%	—	1(+)/9(=)/0(−)	8(+)/2(=)/0(−)
GPSHAP_70%	—	—	6(+)/4(=)/0(−)

evolutionary process on QT-1, QT-3, and QT-4, all methods exhibit similar performance. However, GPSHAP is distinguished by its ability to reduce error more rapidly at the initial phases.

GPSHAP demonstrates a superior ability to enhance the learning and generalization capabilities of GP by efficiently discarding a greater amount of noisy or irrelevant features compared to other feature selection methods. This process enables GP to predominantly utilize relevant features in model construction. A more detailed discussion of the feature selection results facilitated by GPSHAP will be provided in Sect. 6.

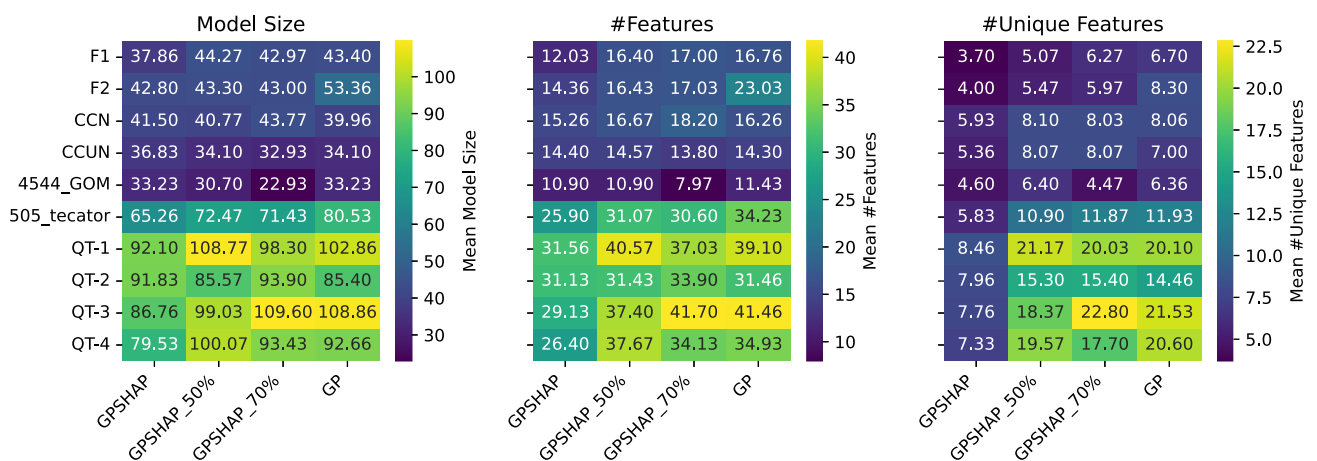


Fig. 9 Comparisons on model size, the number of features and the number of unique features

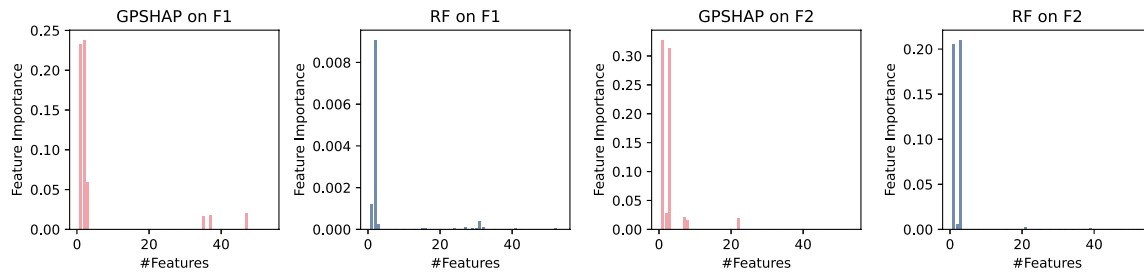


Fig. 10 Feature selection results on the two synthetic datasets

6 Further Analysis

This section delves into an in-depth analysis of the contribution of different components of GPSHAP, feature selection, the regression models generated by GP and the number of model size and features, and computational cost. The aim is to illuminate how GPSHAP surpasses other feature selection methods in enhancing the generalization capabilities of GPSR, offering valuable insights into the mechanisms behind its effectiveness.

6.1 Ablation Analysis

Tables 4 and 5 present the results of statistical significance tests on the NRMSEs of the 30 best-of-run individuals from pairwise comparisons of GPSHAP, GPSHAP_50%, GPSHAP_70%, and GP on all datasets, using the Mann-Whitney U test with a significance level of 0.05. Figure 9 provides an overview of the average model size, the mean number of features, and the number of unique features within these 30 best-of-run models.

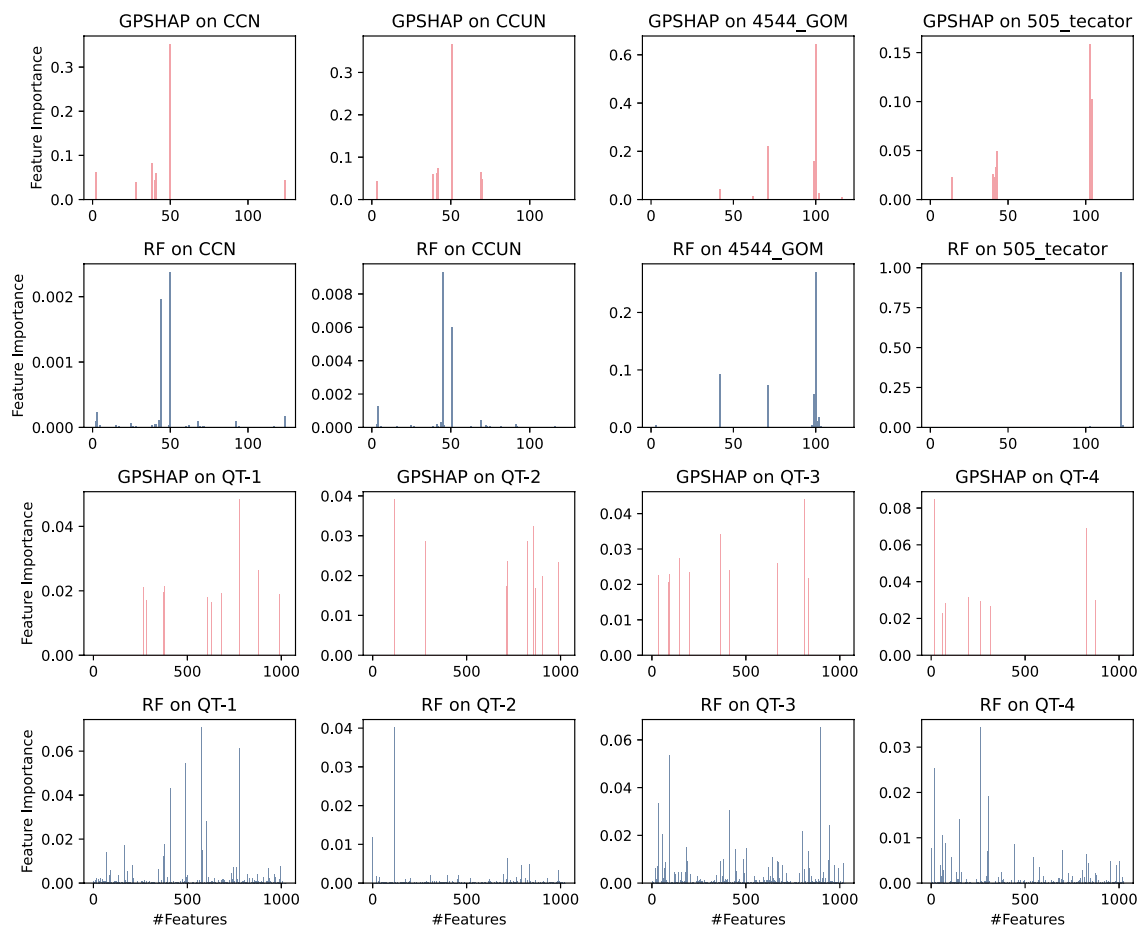


Fig. 11 Feature selection results on the eight real-world datasets

Table 6 Top $\log_2(D)$ features on real-world datasets

Dataset	Method	Top $\log_2(D)$ features
CCN	GPSHAP	50, 38, 2, 41, 124, 40, 28
	RFs	50, 44, 3, 124, 43, 92, 2
CCUN	GPSHAP	51, 42, 69, 41, 39, 70, 3
	RFs	45, 51, 4, 69, 44, 3, 41
4544_GOM	GPSHAP	100, 71, 99, 42, 102, 62, 116
	RFs	100, 42, 71, 99, 102, 101, 98
505_tecator	GPSHAP	103, 104, 43, 42, 40, 14, 41
	RFs	123, 124, 103, 104, 109, 105, 111
QT-1	GPSHAP	778, 876, 376, 265, 374, 681, 993, 608, 281, 627
	RFs	573, 778, 492, 412, 602, 379, 165, 580, 70, 374
QT-2	GPSHAP	115, 855, 281, 826, 719, 988, 822, 903, 714, 868
	RFs	115, 1, 719, 836, 790, 988, 694, 309, 456, 396
QT-3	GPSHAP	814, 366, 145, 669, 412, 199, 92, 37, 833, 89
	RFs	899, 92, 34, 412, 947, 799, 54, 184, 501, 447
QT-4	GPSHAP	20, 826, 197, 875, 265, 78, 318, 199, 58, 62
	RFs	265, 20, 306, 152, 58, 78, 442, 2, 698, 301

When comparing GPSHAP, GPSHAP_50%, and GPSHAP_70% with standard GP, the results highlight the effectiveness of the Shapley value component. GPSHAP, GPSHAP_50%, and GPSHAP_70% all significantly outperform GP on both the training and test sets, indicating that the Shapley value enhances the learning and generalization abilities of GP.

When comparing GPSHAP with GPSHAP_50% and GPSHAP_70%, the results emphasize the importance of the threshold component. GPSHAP performs worse than GPSHAP_50% on the training set, but they exhibit similar performance on the test set. Moreover, although GPSHAP and GPSHAP_70% perform similarly on the training sets, GPSHAP shows better results on the test set. As seen in Fig. 9, GPSHAP produces models with a smaller size, fewer features, and fewer unique features compared to GPSHAP_50% and GPSHAP_70%. This suggests that selecting the top $\log_2(D)$ important features results in simpler models compared with selecting the top 50% or 70% of features.

6.2 Analysis of Feature Importance

The experiment primarily contrasts RF and GPSHAP, as they both exhibit commendable performance on benchmark problems while utilizing different mechanisms, i.e., Shapley value and permutation importance, to assess feature importance. Moreover, the distinction between these

methods resides in their approach to feature searching and tree construction.

Feature importance plays a pivotal role in discerning which features significantly impact the performance of a regression model. GPSHAP uses Shapley value to compute the feature importance, where higher values indicating greater importance of the feature. Since GPSHAP selects only the top $\log_2(D)$ features, this section presents feature importance related to these top $\log_2(D)$ features only. Besides, RF assigns a positive importance value to a feature when the shuffling that feature leads to an increase in regression error, indicating its importance in minimizing the regression error. Essentially, a positive importance value implies that the feature has a beneficial impact on reducing regression error, with larger values denoting greater importance to the target variable. Since GPRF selects only features with positive importance, our analysis will concentrate on the feature importance of these features exclusively.

Figure 10 illustrates the feature importance values determined by GPSHAP and RF on synthetic datasets. On F_1 , both methods identify a few selected features, underscoring their efficacy in significantly reducing the number of features. Notably the three most significant features identified by both methods align with the truly important features, indicating a precise identification of key features. On F_2 , as illustrated in the right part of Fig. 10, a similar pattern to F_1 emerges. Both GPSHAP and RF successfully pinpoint the truly important features, with these features generally exhibiting markedly higher importance values than the rest. This suggests that Shapley value and permutation importance are adept at recognizing noise features.

For the real-world datasets, which possess a higher feature dimensionality compared to the synthetic datasets, a distinct overall pattern emerges. Consider the substantially larger number of features and greater complexity, we specifically examine the detailed importance values for the eight real-world datasets. As shown in Fig. 11, both GPSHAP and RF effectively eliminate a substantial amount of noisy/irrelevant features. The number of features selected by the two methods, however, diverges significantly with GPSHAP selecting a considerably smaller subset of features except on 4544_GOM and 505_tecator. This divergence can be partially attributable to the larger number of trees in RF, which can lead to the inclusion of more redundant features. Additionally, RF's random feature selection scheme allows for a higher likelihood of falsely identifying irrelevant features as important, which would typically be recognized as unimportant and excluded by GP. In summary, when faced with high-dimensional real-world datasets, GPSHAP demonstrates an enhanced capability in selecting the concise important features.

Table 6 arranges the top $\log_2(D)$ features identified by both methods on the eight real-world datasets, ranked by

Table 7 Examples of GP individuals on F_1 , where $F_1 = -g \frac{X_1 X_2}{X_3^2}$ and $g = 6.67E - 11$

Method	Example of best-of-run individual	Simplified individual
GPSHAP	$* (* (* (-1, X_2), \text{sqrt}(\text{inv}(* (X_3, X_3))))), X_1, \text{inv}(X_5))$	$-\frac{X_1 X_2}{X_3^2}$
GP	$\text{inv}(-(+ (X_2, +(-(+ (X_{25}, \text{sqrt}(X_{31}))), +(* (X_3, \text{inv}(X_1))), X_{59}))), \text{sqrt}(\text{sqrt}(-(\text{inv}(X_2), * (X_{49}, X_{14}))))),$ $+ (+ (* (X_3, * (X_3, \text{inv}(X_1))), X_3), +(\text{sqrt}(X_{25}))), -(*$ $(-\text{inv}(X_2), * (X_{49}, X_{14}))), X_3, * (X_{49}, X_{31})),$ $+ (X_2, X_2))))$	$\frac{1}{1/((\frac{3X_2+1}{X_2} + X_{25} - \sqrt{X_{25}} - X_3(\frac{X_3-1}{X_1} + \frac{1}{X_2} - X_{14}X_{49})) - (X_{14}X_{49})\frac{1}{4} + \sqrt{X_{31}} + X_{49}(X_{31}-1))}$
GPPI	$* (* (* (-(- (X_1, 0), * (* (\text{sqrt}(X_1), (- (+ (\text{sqrt}(* (- (X_1, X_{28}), X_{17}), 0))), X_2)), +(* (* (\text{sqrt}(\text{sqrt}(* (X_{32}, X_{35}))), * (\text{sqrt}(-(X_1, 0)), -(X_{32}, 0))))), \text{sqrt}(-(X_1, 0))), X_{28}), -1))$	$\frac{1}{(((X_1 - X_{28})X_{28})^{\frac{1}{2}} + X_{17})^{\frac{1}{2}} * X_1^{\frac{3}{2}} X_2 * (X_1^{\frac{5}{2}} X_8 X_8 X_8 X_{32} X_{35} X_{28} - 1)}$
GPFR	$* (* (* (-(* (* (\text{sqrt}(\text{sqrt}(-(X_{12}, X_2))), \text{sqrt}(* (\text{sqrt}(* (X_5, X_2)), \text{sqrt}(-(X_{12}, X_2))))), X_{13}), -(X_{12}, X_1)), X_1, * (\text{sqrt}(\text{sqrt}(X_5)), X_2))$	$\frac{1}{(((X_1 X_5)^{\frac{1}{4}} (X_{12} - X_2)^{\frac{1}{2}} - X_{13})(X_{12} - X_1 - X_1) * X_1 X_2 X_4^4 - X_1^{\frac{1}{2}} X_2^{\frac{1}{2}} (X_5 X_7)^{\frac{1}{2}} X_6^{\frac{-1}{2}} * (X_6 - X_8)^{\frac{1}{4}} (X_7 - X_8)^{\frac{3}{8}})}$
GPWFS	$* (* (X_1, -1), * (* (X_5, X_7), * (* (\text{sqrt}(\text{sqrt}(* (* (X_2, \text{sqrt}(-(X_7, X_8))), \text{inv}(\text{sqrt}(X_6))))), \text{sqrt}(\text{sqrt}(* (- (X_6, X_8), \text{inv}(* (X_5, X_7))))), \text{sqrt}(\text{sqrt}(* (- (X_7, X_8), \text{inv}(* (X_5, X_7))))))$	

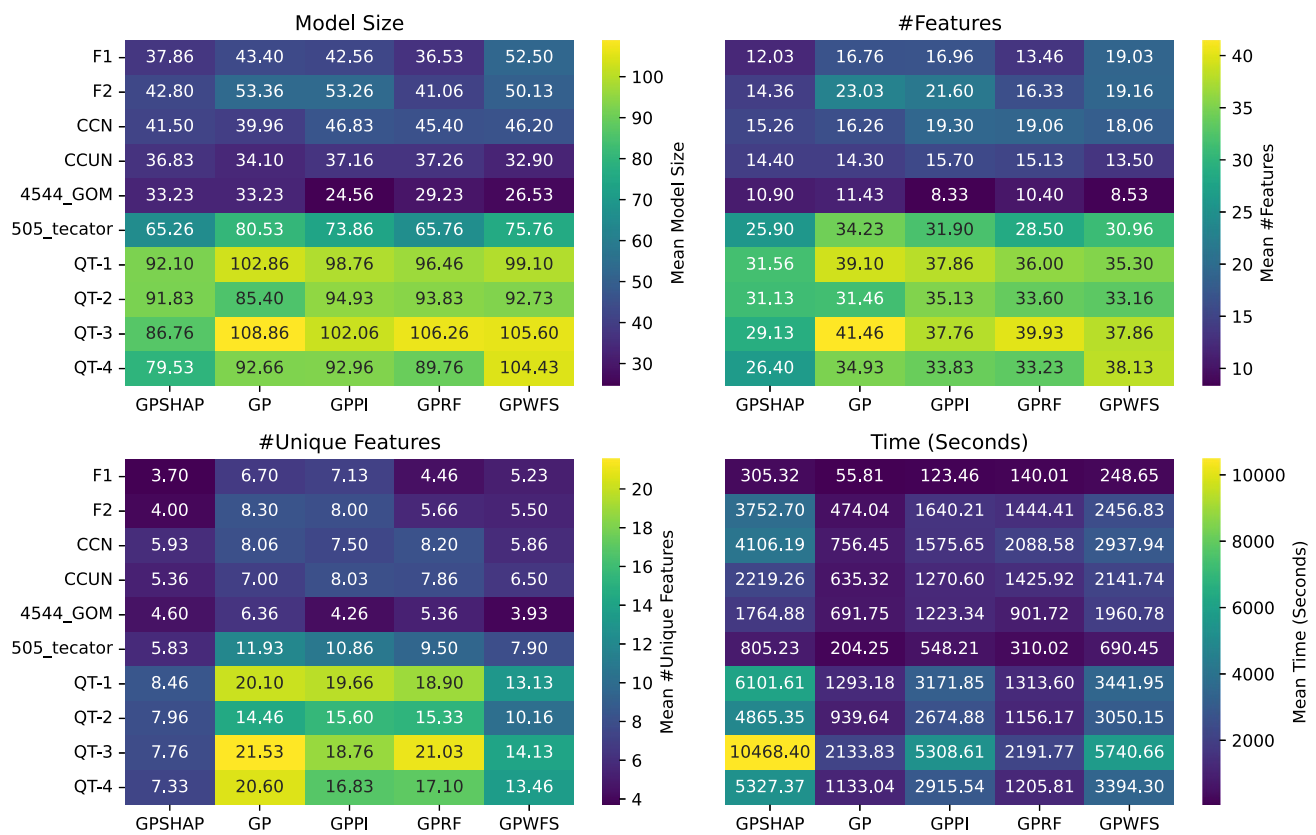


Fig. 12 Comparisons on model size, the number of features, the number of unique features and computational cost (seconds)

their descending importance. Notably, there is a total disparity in the top features pinpointed by the two methods on 505_tecator, QT-1, QT-2, QT-3 and QT-4. This discrepancy in feature selection between the two methods largely stems from the low instance-to-feature ratio. Additionally, the prevalence of irrelevant features also contributes to this inconsistency. In contrast, on the other three datasets, the methods demonstrate more alignment in their selection of important features. Specifically, both methods exhibit similarities in approximately half of the top important features identified. In summary, the instance-to-feature ratio plays a significant role in shaping the feature selection outcomes of GPSHAP and RF. Conversely, for datasets characterized by a higher number of features but fewer instances, the feature selection process proves to be far more challenging.

6.3 Analysis of Evolved Models

Further analysis is conducted on the models created by different methods. We conduct a further analysis on five best models on F_1 , where the true function is known, and all feature selection methods perform effectively. These models are

shown in Table 7. The simplified versions of these models make it easier to analyze the results.

When analyzing important features in the target function $F_1 = -g \frac{X_1 X_2}{X_3^2}$ ($g = 6.67E - 11$), GPSHAP includes all true features and exclusively utilizes the important ones to build models. In contrast, the other three GP-based feature selection methods and standard GP sometimes include extra, unnecessary features or miss some important ones. Model structure further shows that the model generated by GPSHAP more closely resemble the target model compared to those generated by the other methods. Notably, GPSHAP identifies the nearly exact model after 50 generations, which shows it is good at finding the key patterns of the data to generate accurate models.

6.4 Results on Model Size, Number of Features, Number of Unique Features, and Computational Cost

Figure 12 presents an overview of the average model size of the 30 best-of-run models, the mean number of features and unique features within these models, and the average computational cost. The number of features refers to the total count

of all features included in a GP model, while the number of unique features represents the unique features in the model, excluding duplicates. The model sizes in GPSHAP generally appear smaller on 505_tecator, QT-1, QT-3 and QT-4, compared to those in the other four methods, signaling its proficiency in developing more compact models. Although the model size in GPSHAP is slightly larger than those of the other methods on F_1 , F_2 , CCN and QT-2, it incorporates fewer features. This efficiency is likely due to the feature selection mechanism prioritizing the top $\log_2(D)$ features, thereby granting important features enhanced chances to be repeatedly leveraged in constructing models.

Regarding the number of features and unique features, all methods utilize only a fraction of the available features on the datasets. Nonetheless, on eight out of the ten datasets, GPSHAP utilizes fewer features and unique features compared to the other four methods, highlighting the exceptional ability of the Shapley value in identifying crucial features. By effectively filtering out more noisy or irrelevant features while preserving the important ones, GPSHAP enhances model efficiency and generalization performance.

In terms of computational cost, GPSHAP incurs higher expenses than GP due to the additional demands of feature selection, including the computation of Shapley values, gathering features from the best-of-run models, and feature ranking. GPSHAP also proves more time-consuming than GPPI, GPWSF and GPRF due to the higher computational requirements for calculating Shapley values compared to permutation importance, collecting feature frequency and random forests. Despite the increased computational overhead, GPSHAP significantly enhances learning and generalization capabilities and minimizes the feature subset required for model construction, thereby generating more concise models.

7 Conclusions

This paper introduced GPSHAP, a new feature selection method, based on the Shapley value for GPSR, designed to enhance the generalization capabilities of high-dimensional GPSR. Experimental results show that GPSHAP demonstrates superior learning ability compared with standard GP on 7 out of the 10 datasets and significantly better generalization on 6 out of the 10 datasets. GPSHAP consistently outperforms other GP-based feature selection methods in both the learning and generalization on most datasets. Ablation analysis reveals that the Shapley value primarily enhances both training and test performance, while the strategy of selecting the top $\log_2(D)$ feature effectively reduces both the number of unique features and overall model size. Additionally, GPSHAP is more efficient at selecting important features than random forests and excels in developing

models that incorporate only the true features. Furthermore, GPSHAP generates more compact models by selecting fewer features than other methods, achieving at least a 27% reduction in the number of selected features compared with standard GP.

In future work, we aim to enhance the effectiveness of the proposed Shapley value-based feature selection approach by integrating it directly into the evolutionary process, rather than applying it as a separate stage. To further reduce the computational cost of computing the Shapley value, we plan to explore simpler surrogate models for evaluation. Additionally, we intend to extend the proposed Shapley value-based feature selection method to classification tasks, using it to identify the features that most significantly influence class predictions.

Acknowledgements This work is supported by the Marsden Fund of New Zealand Government under Contracts VUW1913, VUW1914, VUW2016, the Science for Technological Innovation Challenge (SfTI) fund under grant E3603/2903, MBIE Data Science SSIF Fund under the contract RTVU1914, and MBIE Endeavor Research Programme under contracts C11X2001 and UOCX2104.

Author Contributions Chunyu Wang is the first author. She primarily drafted the initial manuscript, conducted all experiments, and developed preliminary ideas. Qi Chen is the corresponding author and the second author. She mainly revised the manuscript and contributed to the discussion of ideas. Bing Xue is the third author. She primarily revised the manuscript and contributed to the discussion of ideas. Mengjie Zhang is the last author. He primarily revised the manuscript and contributed to the discussion of ideas.

Funding The Marsden Fund of New Zealand Government under Contracts VUW1913, VUW1914 and VUW2016.

Availability of data and materials All data can be downloaded from OpenML at <https://www.openml.org/>.

Declarations

Conflict of interest Not applicable

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